HARD · ADVANCED MATH

Nonlinear functions

07

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

The function f is defined by $fx = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = fx$ in the xy-plane passes through the points $7, 0$ and $-3, 0$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

A. -6

B. -3

C. 4

D. 5

Q2 ADVANCED MATH

$f(x) = -500x^2 + 25{,}000x$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy-plane intersects the x-axis at 0 and a. What does a represent?

A. The revenue, in dollars, when the unit price of the product is $0

B. The unit price, in dollars, of the product that will result in maximum revenue

C. The unit price, in dollars, of the product that will result in a revenue of $0

D. The maximum revenue, in dollars, that the company can make

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and

P_0

is the population when

$y = 0$

.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$

Q4 GRID-IN ADVANCED MATH

$$h(x) = x^3 + ax^2 + bx + c$$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5, 6, and 7, what is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN ADVANCED MATH

Function f is defined by $f x = -a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = f x - 15$ has a y-intercept at $0, -\frac{99}{7}$. The product of a and b is $\frac{65}{7}$. What is the value of a?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ADVANCED MATH

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $gx = 181.161.4^{x+2}$
II. $hx = 181.4^{x+4}$

A. I only

B. II only

C. I and II

D. Neither I nor II

Q7 ADVANCED MATH

What is the minimum value of the function f defined by $f(x)=(x-2)^2-4$?

A. -4

B. -2

C. 2

D. 4

Q8 ADVANCED MATH

Two variables, x and y, are related such that for each increase of 1 in the value of x, the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A. $y = 4x^{200}$

B. $y = 4200^{x}$

C. $y = 200x^{4}$

D. $y = 2004^{x}$

Q9 ADVANCED MATH

$$fx = 9{,}0000.66^x$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since 1997, and $0 \leq x \leq 5$. If $y = fx$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

A. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 1,708.

B. The minimum estimated number of advertisements the company sent to its clients during the 5 years was 9,000.

C. The estimated number of advertisements the company sent to its clients in 1997 was 1,708.

D. The estimated number of advertisements the company sent to its clients in 1997 was 9,000.

Q10 ADVANCED MATH

$$gx = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the xy-plane, the graph of $y = gx$ passes through the point 022, and $g\text{-}22 = 0$. What is the value of b?

A. 23

B. 22

C. -22

D. -23